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Preliminary Verification of Fixed-Source Sensitivity Analysis in OpenMC

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Abstract — *Sensitivity analysis capabilities have yet to find extensive use in fusion reactor design applications where they can help understand the impact of nuclear data uncertainties on the tritium breeding ratio (TBR), shutdown dose rates, and nuclear heating. Significant uncertainty exists in nuclear data for fusion applications, and the goal of this work is to explore whether adjoint- and perturbation theory–based eigenvalue and generalized response sensitivity methods recently developed within the OpenMC Monte Carlo code can be extended to fixed-source Monte Carlo radiation transport simulations.*

This paper presents a derivation for the extended sensitivity analysis method. The adjoint-based sensitivity coefficients are compared with Monte Carlo SERPENT sensitivity coefficients for a TBR calculation in a simplified ARC-class tokamak. Further verification with the Monte Carlo MCSEN and deterministic ASUSD sensitivity coefficients for the tritium production rate in the Frascati neutron generator helium-cooled pebble bed test blanket module mock-up experiment was performed. The OpenMC sensitivity coefficients were found to agree with the other code systems. The use of the sensitivity coefficients with nuclear data covariance within the sandwich rule for cross-section uncertainty propagation also showed good agreement with the reference total Monte Carlo nuclear data uncertainty.

Keywords — *Fixed source, OpenMC, sensitivity analysis.*

I. INTRODUCTION

Adjoint-weighted quantities have been employed in the open-source Monte Carlo code OpenMC for eigenvalue sensitivity to nuclear data [1] and the calculation of reactor kinetics parameters [2]. They have also been developed and implemented in other Monte Carlo transport codes, such as MCNP [3], TSUNAMI-3D [4], and SERPENT [5]. Sensitivity analysis methods have been used extensively in criticality applications [6] to investigate the impact of nuclear data uncertainties on eigenvalue estimates and to propagate these uncertainties to the responses of interest. However, these methods have found limited use in fixed-source problems [7–11].

These classes of problems occur in fusion reactor design applications, where it is important to evaluate the sensitivity of the tritium breeding ratio (TBR), shutdown dose rates (SDRs), nuclear heating, and energy deposition to nuclear data. Significant uncertainty exists in the nuclear data, design parameters, and operating conditions for fusion reactors, and the effects of these on fusion reactor designs must be known to ensure that the designs meet regulatory and licensing requirements. Analysis of the uncertainties is also important to ensure that the computer codes employed in design simulations are accurate and reliable.

Previous work developed adjoint- and generalized perturbation theory (GPT)–based eigenvalue and generalized response sensitivity analysis methods within OpenMC for fission reactors [12]. This work explores

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the potential of extending the previous work to fixed-source Monte Carlo radiation transport simulations. Within OpenMC, the applications in which the present work enabled the computation of previously unobtainable sensitivity coefficients are in fusion reactor designs. These include the computation of the sensitivities of the TBR and tritium production rates in breeding blanket materials and of neutron heating in structural materials.

The work presented here provides the foundation for future sensitivity analyses of OpenMC-coupled transport simulations. The adjoint flux, i.e., the particle's importance, which is needed in the sensitivity coefficient calculation, is determined in the forward simulation based on the CLUTCH (contributon-linked eigenvalue sensitivity/uncertainty estimation via track length importance characterization) method [4]. The sensitivity coefficients are derived from first-order perturbation theory applied to the forward and adjoint radiation transport equations.

To verify the sensitivity coefficients, this work compares the sensitivities calculated by OpenMC, SERPENT version 2.2.1, and MCSEN Monte Carlo, and the deterministic ASUSD for a simplified ARC-class tokamak [13] and the Frascati neutron generator (FNG) helium-cooled pebble bed (HCPB) test blanket module (TBM) mock-up experiment [14]. The combination of the sensitivity coefficients and nuclear reaction cross-section covariance in the sandwich rule for uncertainty propagation yielded TBR and tritium production rate nuclear data uncertainties that were in good agreement with the reference total Monte Carlo (TMC) method [15,16].

II. THEORY

The relative sensitivity coefficient is defined as the ratio of the fractional change in the response R to the fractional change in input parameter α,

$$S = \frac{\partial R/R}{\partial \alpha/\alpha} \ . \tag{1}$$

In this work, the responses included the particle flux and absolute reaction rates (responses that are linear functionals of the flux), and the input parameter represents nuclear data. The response can be written as

$$R = \langle f_R, \psi \rangle \, , \tag{2}$$

where f_R is a response function, e.g., reaction cross section, ψ is the forward angular flux, and the angle brackets denote integration over all phase space (position, direction, and energy). The sensitivity coefficient can be expressed using derivatives of the input parameter α as

$$S_{R,\alpha} = \frac{\alpha}{R}\left[\left\langle \frac{\partial f_R}{\partial \alpha}, \psi \right\rangle + \left\langle f_R, \frac{\partial \psi}{\partial \alpha} \right\rangle\right] \ , \tag{3}$$

where the first and second terms are generally referred to as the direct and indirect effects, respectively. The direct effect, when multiplied by α, is the reaction rate if f_R is a function of α and zero otherwise. The indirect effect requires the use of perturbation theory and the forward and adjoint transport equations because it captures how α impacts the global solution and its influence on R.

The forward and adjoint transport equations that govern the balance of the neutron flux in a system subject to an external source Q can be given, respectively, as

$$H\psi = Q \tag{4}$$

and

$$H^\dagger \psi^\dagger = S^\dagger = f_R \ , \tag{5}$$

where H is the forward transport operator, $H^\dagger$ is its adjoint, Q is the external source, and $S^\dagger$ is the adjoint source. The adjoint source $S^\dagger$ is defined in a way that enables the computation of the indirect term,

$$S^\dagger = f_R \ . \tag{6}$$

Multiplying the derivative of the forward equation by the adjoint function, and integrating over all phase space, results in

$$\left\langle \psi^\dagger, H\frac{\partial \psi}{\partial \alpha} \right\rangle = \left\langle \psi^\dagger, \frac{\partial Q}{\partial \alpha} \right\rangle - \left\langle \psi^\dagger, \frac{\partial H}{\partial \alpha}\psi \right\rangle \ . \tag{7}$$

Applying the property of the adjoint to the left side of Eq. (7), and substituting from Eq. (5), we have an expression for the indirect term in Eq. (3),

$$\left\langle \psi^\dagger, H\frac{\partial \psi}{\partial \alpha} \right\rangle = \left\langle \frac{\partial \psi}{\partial \alpha}, H^\dagger \psi^\dagger \right\rangle = \left\langle f_R, \frac{\partial \psi}{\partial \alpha} \right\rangle \ . \tag{8}$$

By substituting Eq. (8) into Eq. (3), we have

$$S_{R,\alpha} = \frac{\alpha}{R} \left[\left\langle \frac{\partial f_R}{\partial \alpha}, \psi \right\rangle + \left\langle \psi^\dagger, \frac{\partial Q}{\partial \alpha} \right\rangle - \left\langle \psi^\dagger, \frac{\partial H}{\partial \alpha}\psi \right\rangle\right] \ . \tag{9}$$

Eq. (9) can be rewritten to give the relative sensitivity coefficient as [17]

$$S_{R,\alpha} = \frac{\langle f_{R,\alpha}, \psi \rangle + \left\langle \psi^\dagger, Q_\alpha \right\rangle - \left\langle \psi^\dagger, H_\alpha \psi \right\rangle}{R} , \tag{10}$$

where $f_{R,\alpha}$ is the portion of the response function that depends on nuclear data α, $\psi^\dagger$ is the adjoint angular flux, Q_α is the portion of the source term impacted by the nuclear data α; and H_α is the portion of the collision, scattering, and fission transport operators depending on nuclear data α.

II.A. Contributon Theory for Fixed-Source Radiation Transport Simulations

The adjoint flux can be computed from the contributon theory [18] which has been adapted to continuous-energy Monte Carlo eigenvalue simulations in the CLUTCH method [4]. To compute the adjoint flux in an eigenvalue sensitivity analysis, contributon theory equates all the neutron sources to the fission source weighted by the system eigenvalue, i.e.,

$$Q = \lambda F \psi . \tag{11}$$

Taking the inner product of Eq. (11) with the adjoint flux,

$$\left\langle \psi^\dagger, Q \right\rangle = \left\langle \psi^\dagger, \lambda F \psi \right\rangle . \tag{12}$$

Consider a single neutron traveling in phase space τ, such that

$$Q = Q_o \delta(\tau) . \tag{13}$$

Eqs. (12) and (13) can be expressed as the Green's function,

$$\psi^\dagger(\tau) = \frac{\lambda}{Q_o} \psi^\dagger(r) F(r) \psi(\tau \rightarrow r) . \tag{14}$$

Eq. (14) states that the importance of the particle traveling in phase space τ is determined by the number of fission neutrons it creates at location r after leaving τ, weighted by the importance of the daughter fission neutrons.

To adapt the existing contributon theory to fixed-source radiation transport, a similar process is done for the neutron balance equations in Eqs. (4) and (5). Taking the inner product of the forward equation weighted by the adjoint flux and the inner product of the adjoint equation weighted by the forward flux,

$$\left\langle \psi^\dagger, H\psi \right\rangle = \left\langle \psi^\dagger, Q \right\rangle \tag{15}$$

and

$$\left\langle \psi, H^\dagger \psi^\dagger \right\rangle = \left\langle \psi, S^\dagger \right\rangle . \tag{16}$$

Applying the property of the adjoint [the left side of Eqs. (15) and (16) are identical] gives

$$\left\langle \psi^\dagger, Q \right\rangle = \left\langle \psi, S^\dagger \right\rangle . \tag{17}$$

Again, considering a neutron source of one single particle traveling in phase space such as in Eq. (13), we can compute the adjoint neutron flux from the Green's function,

$$\psi^\dagger(\tau) = \frac{1}{Q_o} G(\tau \rightarrow r) S^\dagger dr , \tag{18}$$

where $G(\tau \rightarrow r)$ is a transfer function which is equal to the number of particles produced at all energies and directions at r due to a source particle born at τ. The adjoint flux calculation from the CLUTCH method then estimates the importance of a particle $\psi^\dagger(\tau)$ as how much the particle contributes to some response of interest. For the particle born in τ, the flux generated in the response region r weighted by the adjoint source $S^\dagger$ is given in Eq. (19) [10] as the adjoint flux,

$$\psi^\dagger(\tau) = \psi(\tau \rightarrow r) S^\dagger . \tag{19}$$

The adjoint source is the detector response function in the forward simulation. Eq. (19) can be evaluated in an OpenMC forward simulation as the weighted flux of the particle, as it scores in the detector region. The adjoint flux is used in Eq. (10) within the same OpenMC simulation to tally the sensitivity coefficients.

II.B. Implementation

Eigenvalue and generalized response (e.g., reaction rate ratios) sensitivity analyses have been implemented in OpenMC. In prior work, the adjoint flux for criticality problems in OpenMC was calculated in a separate run using the fission matrix approach to account for the fission multiplication component of the adjoint flux.

Details of this can be found in Ref. [12] and are not repeated here.

OpenMC was modified to incorporate sensitivity analysis in the fixed-source mode for responses such as fluxes and reaction rates. The adjoint flux in fixed-source mode, however, is calculated during the same run as the sensitivity coefficients. During the transport of a particle, the $H_\alpha\psi$ term is calculated for each nuclear data whose perturbation is considered. This is then multiplied by the importance of that particle. This product is accumulated for every particle in each batch and each response of interest in the sensitivity analysis.

The first term in Eq. (10) is the response reaction rate from α, the second term is the adjoint-weighted source from α, and the third term is the adjoint-weighted interaction rate, scattering, and fission sources from α. The first term (or direct term) is only calculated if, for example, f_R is the total tritium production rate and α is the ^{6}Li microscopic tritium production cross section. The second term is the adjoint-weighted source, which accounts for physical events of various nuclides that could impact the source, and there was no such dependence in this work.

For the problems that would be considered in fusion applications, the adjoint-weighted transport [third term in Eq. (10)] is split into terms representing scattering and total sensitivities computed in OpenMC according to Eqs. (20) and (21), respectively,

$$S_{\text{scatter}} = \sum_{n\in G}\sum_{k\in {}^nC^i}\frac{\partial\Sigma_s(r_k,E_k)}{\partial\alpha}\psi^\dagger \tag{20}$$

and

$$S_{\text{total}} = -\sum_{n\in G}\sum_{t\in T_n^i} l_t\frac{\partial\Sigma_t(r_t,E_t)}{\partial\alpha}\psi^\dagger\ , \tag{21}$$

where G is the neutron generation, ${}^nC^i$ is the set of collisions that particle n undergoes up to and including collision i, r_k is the location precollision k, E_k is the energy precollision k, t is the neutron track, T is the neutron path, and l_t is the track length.

The spatial dependence of the adjoint flux has been omitted in Eqs. (20) and (21) for the sake of conciseness. Please note that this does not mean or assume that the adjoint flux is constant for each neutron generation. The terms in Eqs. (20) and (21) represent the derivative of the scattering cross section and total reaction rate with respect to the nuclear data α, respectively. These are based on the same routines already implemented in OpenMC for eigenvalue sensitivity. More information on the implementation can be found in Ref. [12].

III. RESULTS

This section presents and discusses the results from two different problems that enabled us to assess the accuracy of the fixed-source CLUTCH sensitivity implementation in OpenMC. All the results are based on calculations using a modified version of OpenMC version 0.15.2 branched from Ref. [19]. First, a simplified tokamak consisting of a molten salt tritium breeding blanket is examined in order to analyze the sensitivity of the TBR to the nuclear data of materials present in the blanket. The results are compared to SERPENT Monte Carlo results. Next, we apply the sensitivity analysis to the FNG HCPB benchmark problem [14] from the Shielding INtegral Benchmark Archive and Database (SINBAD) [20], and the OpenMC results are compared against both the ASUSD and MCSEN results reported in the literature.

III.A. Simplified ARC-Class Tokamak

This is a simplified version of the ARC-class tokamak first developed by the Massachusetts Institute of Technology and Commonwealth Fusion Systems [13]. This is shown in Fig. 1. The tokamak employs a liquid immersion blanket for tritium breeding. The simplified reactor was modeled as a 300-cm-radius spherical geometry without shielding, as shown in Fig. 2, with a 14-MeV isotropic point source to represent the neutrons from the deuterium-tritium reaction. A molten salt tritium breeding blanket made of ClLif (LiCl + LiF) (TBR = 0.9967) with natural lithium was used.

The response of interest for which the sensitivities were sought was the TBR of the blanket. The TBR can be calculated within OpenMC using the transport cross sections of the (n,Xt) reactions of the tritium-producing nuclides ^{6,7}Li. It can also be calculated using activation cross sections from the International Reactor Dosimetry and Fusion File library [21] as the energy multiplier. The tritium production cross section was the adjoint source during the adjoint flux calculation.

Energy-dependent sensitivity profiles were calculated in the VITAMIN-J 175 energy group. OpenMC-calculated sensitivity coefficients for the test problem were compared to the Monte Carlo GPT results obtained with SERPENT version 2.2.1. The SERPENT calculations were based on real (accepted) and virtual (rejected) events (interactions) that form the collision history of a particle. The probability

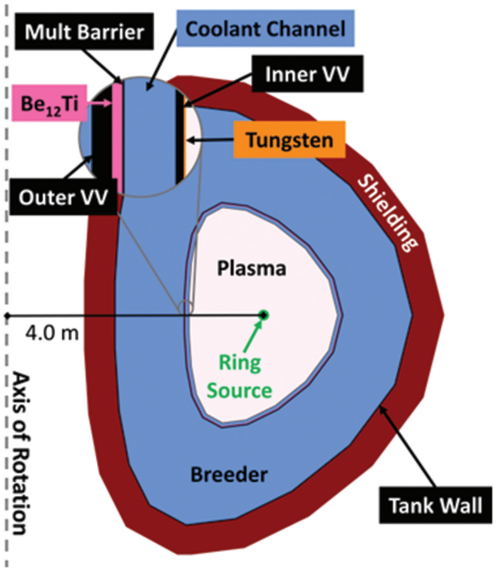


Fig. 1. Cross-sectional view of the simplified tokamak in OpenMC [15].

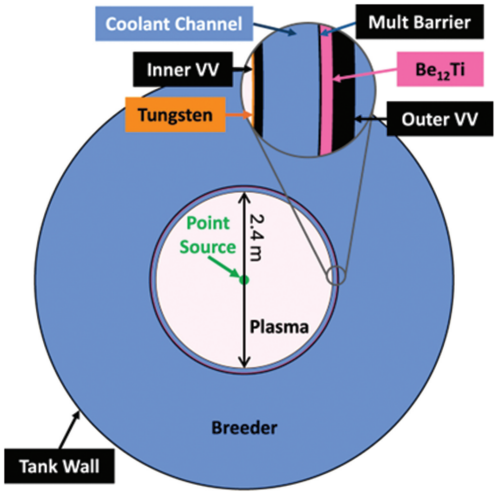


Fig. 2. Cross-sectional view of the spherical model of the simplified tokamak [15].

density functions of the considered events and the sampling process were biased, and fairness was introduced in the transport by rejecting the sampled events based on the biases. The collision history was combined with weight perturbation and used to calculate the effects of perturbation on the response of interest [5].

The neutron flux in the breeding blanket, first wall, inner and outer vacuum vessels (VVs), and multiplier is shown in Fig. 3. Energy-integrated sensitivities (in %/%) of the TBR with respect to the ^{35,37}Cl, ^{19}F, and ^{6,7}Li reaction cross sections are summarized in Table 1 with their relative statistical uncertainties in percent in parentheses. Sensitivities less than or equal to 1.0×10^{-4} are not shown in Table 1.

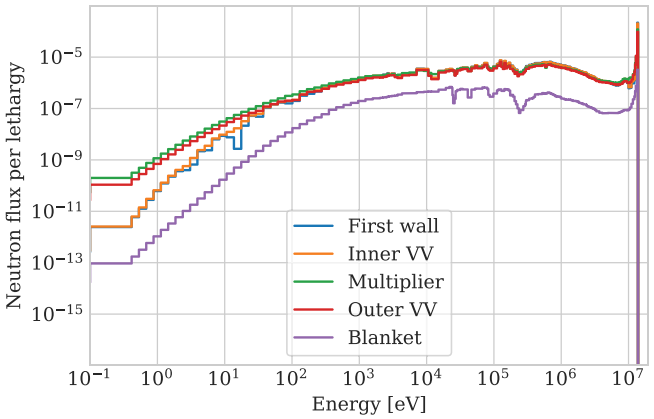


Fig. 3. Neutron flux spectrum in the simplified tokamak.

The table includes the ratio of OpenMC sensitivities to those of SERPENT for verification. The closer the ratio is to 1.0, the better the agreement.

The results were obtained using ENDF/B-VII.1 cross sections. The nuclides were those present in the breeding blanket. The strong negative sensitivity coefficient of the ^{35}Cl (n,p) cross section means that a decrease in the neutron population via parasitic absorption negatively impacts the TBR. The same can be said of the (n,α) cross section. This is why chloride-based materials are proposed to be enriched in ^{37}Cl to reduce parasitic absorption in ^{35}Cl [22]. Enrichment in ^{37}Cl also enhances its neutron multiplication properties. The ^{37}Cl $(n,2n)$ cross section is the highest of its sensitivities, and the positive value confirms that neutron multiplication leads to positive contributions to the TBR.

The TBR in ClLiF is mostly sensitive to the ^{6}Li (n,t), ^{35}Cl (n,p), (n,α), and ^{37}Cl $(n,2n)$ cross sections. It has been noted that ^{35}Cl (n,p) drives the reactivity uncertainty in the molten chloride fast reactor [23], which has necessitated a reevaluation of the cross section [24]. The ^{19}F and ^{7}Li reactions contribute one order of magnitude less sensitivity. Sensitivity to the elastic cross section can be seen to be small, on the order of 10^{-3} in the nuclides.

It is likely that the combined effect of the elastic sensitivities could be responsible for some of the slowing down of the high-energy neutrons to thermal energies for the ^{6}Li (n,t) reaction. Reasonable agreement between OpenMC and SERPENT was observed. The difference in the energy-integrated sensitivity coefficient for most of the reactions was within 10%.

The sensitivity profiles were calculated in the VITAMIN-J 175 group structure. Presented in Fig. 4 are the results for the ^{35}Cl (n,p) cross section. The results were normalized to the lethargy width of the energy bins. The ^{35}Cl (n,α) cross-section sensitivity is not shown because it has the same profile as those of ^{35}Cl (n,p). Around the

TABLE 1

Energy-Integrated Sensitivity Coefficients for the Simplified Tokamak TBR, with Relative Statistical Uncertainties of the Sensitivity Coefficients Shown in Parentheses. These Are Omitted for Uncertainties Less Than 0.1%

Isotope	Reaction	OpenMC (%/%)	SERPENT (%/%)	OpenMC/SERPENT
^{35}Cl	(n,p)	-8.1923×10^{-2}	-1.0318×10^{-1}	0.794
^{35}Cl	(n,α)	-4.4000×10^{-2}	-4.8060×10^{-2}	0.915
^{35}Cl	Elastic	1.3270×10^{-3} (3.2)	2.0086×10^{-3} (4.0)	0.661
^{35}Cl	(n,γ)	-2.1754×10^{-2}	-2.2106×10^{-2}	0.984
^{35}Cl	(n,d)	-5.4587×10^{-3}	-5.9185×10^{-3}	0.922
^{37}Cl	(n,p)	-1.6912×10^{-3}	-1.8734×10^{-3}	0.903
^{37}Cl	(n,α)	-2.8150×10^{-3}	-3.1428×10^{-3}	0.896
^{37}Cl	$(n,2n)$	2.2657×10^{-2} (0.4)	1.9447×10^{-2}	1.165
^{37}Cl	(n,γ)	-1.3785×10^{-3}	-1.3884×10^{-3}	0.993
^{19}F	Elastic	2.1219×10^{-3} (3.1)	3.6250×10^{-3} (2.5)	0.585
^{19}F	$(n,2n)$	3.8453×10^{-3} (0.9)	3.2784×10^{-3}	1.173
^{19}F	(n,α)	-8.8406×10^{-3}	-9.4534×10^{-3}	0.935
^{19}F	(n,p)	-3.4958×10^{-3}	-3.8243×10^{-3}	0.914
^{19}F	(n,d)	-2.5409×10^{-3}	-2.7833×10^{-3}	0.913
^{19}F	(n,t)	-1.3718×10^{-3}	-1.4977×10^{-3}	0.916
^{19}F	(n,γ)	-1.3161×10^{-3}	-1.3310×10^{-3}	0.989
^{6}Li	(n,t)	-6.9598×10^{-1}	-7.0108×10^{-1}	0.993
^{7}Li	Elastic	6.3440×10^{-3} (1.0)	5.5038×10^{-3} (1.5)	1.152
^{7}Li	$(n,2n)$	1.0612×10^{-2} (0.6)	9.1138×10^{-3}	1.164
^{7}Li	(n,d)	-3.2162×10^{-3}	-3.5033×10^{-3}	0.918

14-MeV region, where the D-T neutron peak usually occurs, the dip in the sensitivity coefficient of the ^{35}Cl (n,p) ross -section means that an increase in the neutrons that are absorbed (but not in ^{6}Li) would decrease the TBR. The low-energy region shows a much smaller contribution to the (n,p) sensitivity. This sensitivity profile is similar to the one obtained for the molten chloride fast reactor [23] where the dip occurs between 1 MeV and 10 MeV.

In Fig. 5, TBR sensitivity to the ^{37}Cl $(n,2n)$ is shown. This reaction had the largest sensitivity for ^{37}Cl, as the other reactions had sensitivities with one order of magnitude less. At 14 MeV, the peak in the $(n,2n)$ sensitivity means that an increase in neutron multiplication enhances the TBR. The OpenMC sensitivity profile for ^{37}Cl $(n,2n)$ was higher than SERPENT in the peak region.

In Fig. 6, the TBR sensitivity to the ^{6}Li (n,t) is shown to be negative between 10^{-4} MeV and 14 MeV. This cross section drives the ^{6}Li total cross-section sensitivity, as the other cross sections had sensitivities less

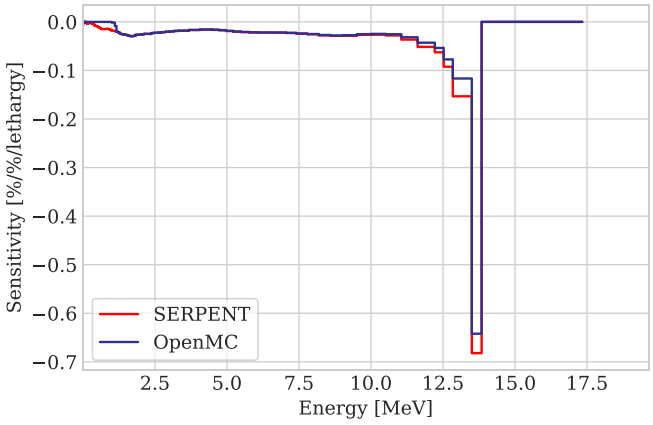


Fig. 4. Comparison of TBR sensitivity to the ^{35}Cl (n,p) cross section in the blanket between OpenMC and SERPENT.

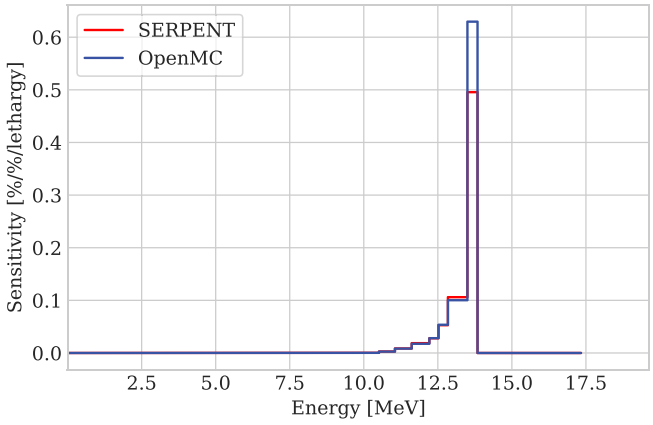


Fig. 5. Comparison of TBR sensitivity to the ^{37}Cl $(n,2n)$ cross section in the blanket between OpenMC and SERPENT.

than 10^{-2}. The negative value was due to the reduction of the neutron flux caused by increased neutron absorption in ^{6}Li. There is a small dip at 14 MeV and a large dip around 10^{-3} MeV. This ^{6}Li sensitivity profile is similar to the one obtained for the European DEMO fusion reactor with the HCPB breeding blanket made of ceramic material (Li_4SO_4), although the main driver there in Ref. [25] was the elastic cross section.

In the sensitivity profiles, a good agreement between the two codes could be observed. The statistical error of the OpenMC energy-dependent sensitivity coefficients in the figures is within 10% in the dip and peak regions and more than 10% in the energy regions with sensitivities close to zero.

The uncertainties of the TBR in the tokamak have been previously analyzed using ENDF/B-VIII.0 cross sections and TENDL-2019 covariances. The same nuclear data were used in this section to calculate the sensitivity profiles and compare the uncertainty propagation results obtained by the TMC method [15] and the sandwich rule in Eq. (22),

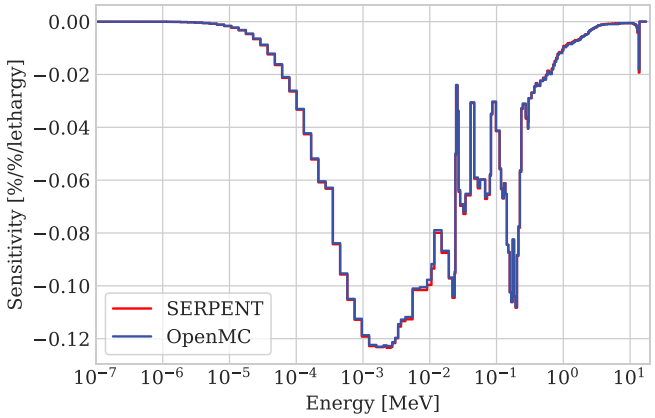


Fig. 6. Comparison of TBR sensitivity to the ^{6}Li (n, t) cross section in the blanket between OpenMC and SERPENT.

$$\sigma_R^2 = S_{R,\alpha}^T C\, S_{R,\alpha}\,. \tag{22}$$

Using the covariance data C in Eq. (22), processed and provided in the same energy group structure as the sensitivity coefficients by the ERRORR module of NJOY [26], the uncertainties of the TBR were calculated. Although the TBR was more sensitive to the ^{6}Li (n, t) reactions, most of the uncertainty came from the ^{35}Cl (n, p) and (n, α) cross sections. The uncertainties calculated by the sensitivity coefficients and TMC approach [15] were about 2% at one sigma standard deviation. The cross-section uncertainty is shown in Table 2, where the "Total" row refers to all the partial/individual reactions, including those not shown but that have covariance available in the nuclear data library.

The TMC results were based on random cross sections from SANDY [27]. SANDY generates random samples of neutron cross sections based on nuclear data and their covariances in the evaluated nuclear data files. Over 300 OpenMC simulations were run, each with different perturbed cross sections to build the random TBRs for statistical analysis of the uncertainties, whereas the uncertainty analysis by the OpenMC adjoint GPT approach requires one simulation to tally the sensitivity coefficients and the uncertainty propagation with covariance is carried out as a post-processing step.

Since the main objective of this work was to verify the accuracy of the sensitivity coefficients, the performance comparison between the adjoint GPT and TMC for uncertainty analysis was not conducted in this paper but left as future work. However, we can still say the adjoint GPT approach is faster than the TMC method. Another advantage of the adjoint GPT approach is the requirement for relatively fewer particle histories, whereas the TMC approach demands multiple runs and larger particle histories to ensure that the statistical uncertainty from the Monte Carlo method is much smaller than that arising from the cross-section perturbations. The statistical error of the sensitivity coefficients is not yet propagated to the nuclear data uncertainties in Table 2

TABLE 2

Nuclear Data Uncertainty of the TBR Calculated by OpenMC due to Chlorine Reaction Cross Sections, with the Adjoint-GPT and SANDY-TMC Approaches Used Separately with OpenMC for the Uncertainty Propagation

Nuclide	Reaction	Adjoint-GPT (%)	SANDY-TMC (%)
^{35}Cl	Total	1.94	2.01
	(n, p)	1.78	1.79
	(n, α)	1.22	1.24
^{37}Cl	Total	0.42	0.51
	$(n, 2n)$	0.40	–

for the adjoint-GPT results. The good agreement between the results shows that the sensitivity coefficients were behaving as expected and can be used for subsequent sensitivity and uncertainty analyses.

III.B. FNG HCPB TBM Problem

The second and final problem that we analyzed was the FNG HCPB benchmark. This a TBM experiment that was performed at ENEA (Italian National Agency for New Technologies, Energy and Sustainable Economic Development) in 2005. The experiment was performed to validate the accuracy of transport codes and nuclear data to predict the tritium production rate in one of the proposed European TBMs with relevant features typical of the TBM planned to be installed in ITER.

The mock-up was made of a stainless steel box, with dimensions 31 × 31 × 29 cm, filled with beryllium blocks and breeder cassettes containing Li_2CO_3 powder with natural lithium enrichment 7.5% ^{6}Li and 92.5% ^{7}Li. The breeder cassettes were located above and below the D-T neutron beam line. The tritium production rate was measured at four depths inside the beryllium block.

One measurement at each depth was performed using Li_2CO_3 pellets in the upper cassette and lower cassette, above and below the D-T neutron beam line, respectively.

The upper cassettes were referred to as stacks 1, 3, 5, and 7, while the lower cassettes were referred to as stack 2, 4, 6, and 8. The FNG HCPB TBM experiment with the location of the stacks is shown in Fig. 7. A 14-MeV FNG neutron source was placed 5.3 cm in front of the block for the irradiation. The rear of the stainless steel box was attached to another stainless steel box with dimensions 31 × 12.7 × 31 cm containing Li_2CO_3 powder. The stainless steel block was 0.5 cm thick.

Nuclear data sensitivity and uncertainty analysis results of the tritium production rates previously obtained using the MCSEN [28] and ASUSD/DENOVO [16] workflows provided data for validating the OpenMC fixed-source sensitivity capability. MCSEN is a local update and patch to MCNP [29]. ASUSD is a program package containing ADVANTG and SUSD3D. SUSD3D [30], a deterministic code based on GPT for sensitivity and uncertainty analyses, is coupled to the DENOVO discrete ordinates code [31], which is part of ADVANTG.

Sensitivities and uncertainties were obtained for the tritium production rate due to ^{6}Li in the pellets at stacks 1, 3, 5, and 7. The energy-integrated sensitivities calculated by OpenMC, which were compared to the ASUSD and MCSEN results, are presented in Table 3. The ASUSD results were based on the ENDF/B-VII.1 nuclear data library. The MCSEN results employ older evaluations. The ratio of OpenMC sensitivities to those of ASUSD is included in the table for verification. The closer the ratio is to 1.0, the better the agreement.

Only the sensitivities due to total, elastic, and $(n, 2n)$ cross sections are shown. The agreement between the codes for the four upper stacks was considered good. The total sensitivity for the (n, p), (n, t), (n, α), and (n, γ) cross sections are available, but are not shown in the table because they are small.

The energy-dependent sensitivity profiles for the elastic and $(n, 2n)$ cross sections are compared in Figs. 8 and 9 for stack 1. The ASUSD results were obtained from Ref. [32]. Except for the slight discrepancies in the peak region, the agreement between the codes is good. The discrepancy is said to be due to the source

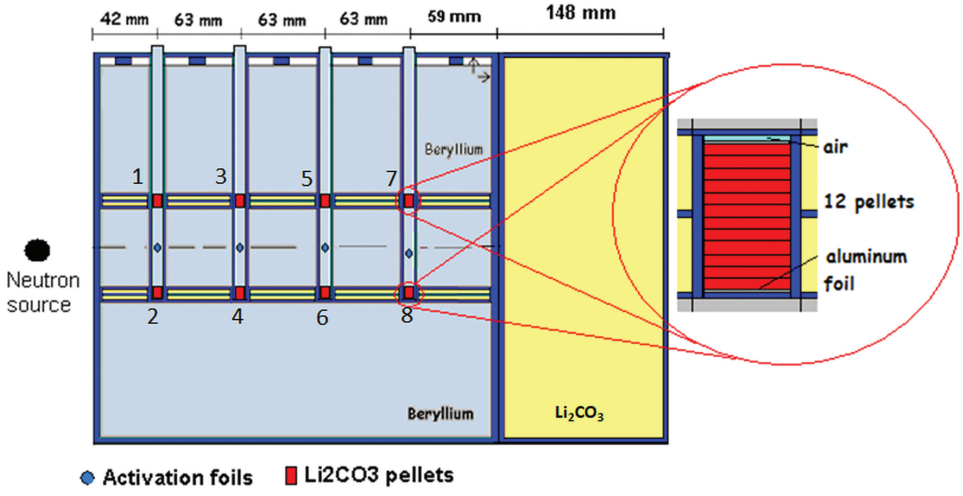


Fig. 7. FNG HCPB TMB mock-up experiment [14].

TABLE 3

Energy-Integrated Sensitivity of ^{6}Li Tritium Production in the Stacks to the ^{9}Be Cross Sections

Stack	Reaction	OpenMC (%/%)	ASUSD (%/%)	MCSEN (%/%)	OpenMC/ASUSD
1	Total	2.63	2.63	2.67	1.000
	Elastic	2.17	1.99	2.00	1.090
	$(n, 2n)$	0.50	0.67	0.72	0.746
3	Total	2.63	2.67	2.75	0.985
	Elastic	2.14	2.06	2.09	1.039
	$(n, 2n)$	0.52	0.66	0.71	0.788
5	Total	2.37	2.28	2.40	1.040
	Elastic	1.88	1.73	1.78	1.087
	$(n, 2n)$	0.54	0.61	0.66	0.885
7	Total	2.01	1.87	2.22	1.075
	Elastic	1.51	1.38	1.67	1.094
	$(n, 2n)$	0.54	0.56	0.61	0.964

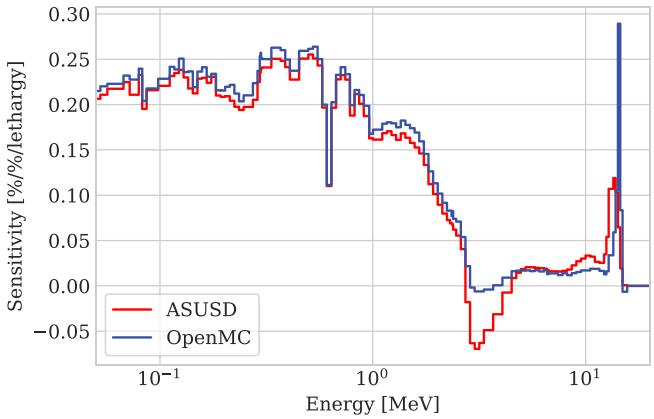


Fig. 8. Comparison of the OpenMC and ASUSD ^{6}Li tritium production sensitivity profiles in stack 1 with respect to the ^{9}Be elastic cross section.

description in DENOVO [16]. The OpenMC-calculated sensitivities for Stacks 1, 3, 5, and 7 are shown in Figs. 10 and 11 for the elastic and $(n, 2n)$ cross sections. Compared to the $(n, 2n)$ cross section, the changes in the sensitivities to the elastic cross section below 1 MeV and around 14 MeV are more visible across the stacks. This is because the neutron spectrum became softer at locations away from the source. Among the upper stacks, stack 1 was closest to the neutron source, as shown in Fig. 7.

Nuclear data uncertainties of the tritium production rate were propagated using ENDF/B-VII.1 covariances with OpenMC sensitivity coefficients for comparison with the ASUSD and TMC results obtained with SANDY in Ref. [16]. The results are presented in Table 4. The uncertainties are given at one sigma

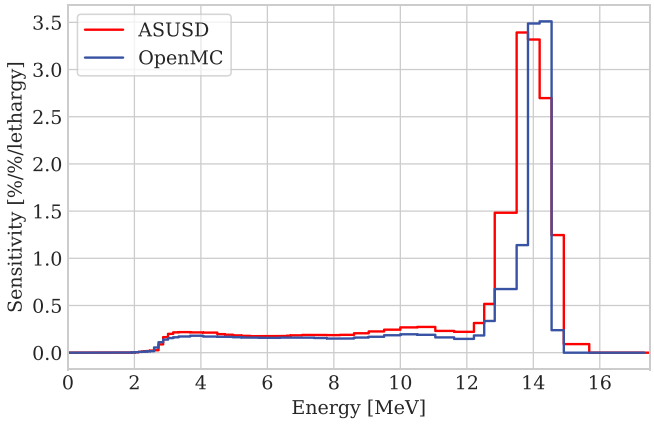


Fig. 9. Comparison of the OpenMC and ASUSD ^{6}Li tritium production sensitivity profiles in stack 1 with respect to the ^{9}Be $(n, 2n)$ cross section.

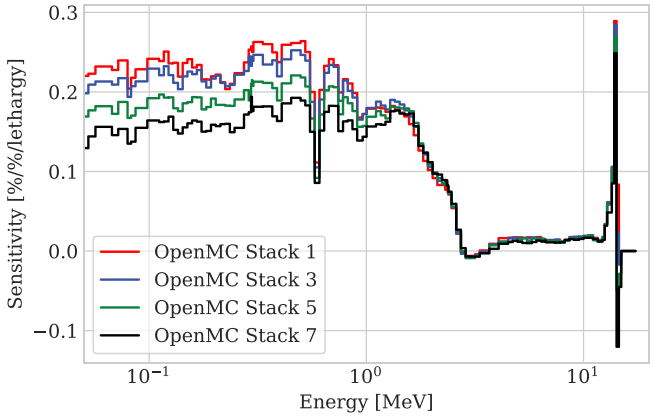


Fig. 10. Comparison of the OpenMC ^{6}Li tritium production sensitivity profiles in stacks 1, 3, 5, and 7 with respect to the ^{9}Be elastic cross section.

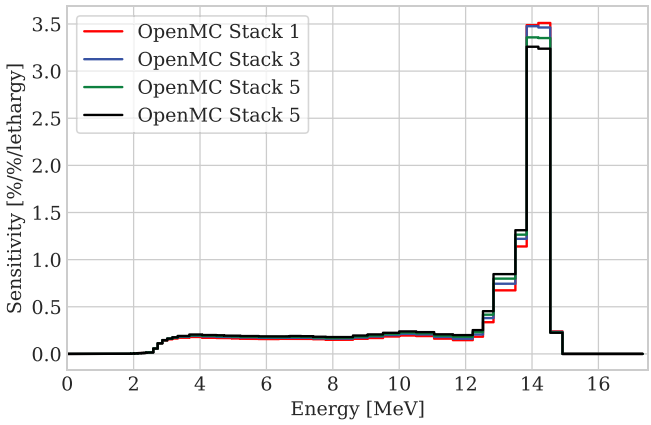


Fig. 11. Comparison of the OpenMC ^{6}Li tritium production sensitivity profiles in stacks 1, 3, 5, and 7 with respect to the ^{9}Be $(n, 2n)$ cross section.

standard deviation. The SANDY results were not available in Ref. [16] for stacks 3 and 5 and for the elastic cross section in stacks 1 and 5.

For the FNG HCPM TBM, the nuclear data uncertainty propagation results showed good agreement, as in the case of the simplified ARC-class fusion device. The nuclear data uncertainty was observed to be lower the farther we are away from the neutron source.

IV. CONCLUSION

This work extended the recently implemented adjoint sensitivity and GPT capabilities in OpenMC [33] to fixed-source Monte Carlo radiation transport problems. OpenMC sensitivity coefficients obtained by the GPT and CLUTCH method were compared against SERPENT Monte Carlo calculations obtained by the collision history method [5], MCSEN code, and deterministic results from SUSD3D. Energy-integrated and energy-dependent sensitivities from the three codes were compared to verify the accuracy of the implementation.

The uncertainties propagated with the OpenMC sensitivity coefficients were further compared to the TMC results. The fixed-source sensitivity analysis within OpenMC was tested on two problems: a simplified tokamak and the FNG HCPB TBM benchmark experiment. The results for the two problems showed that the overall agreement of the sensitivity coefficients and nuclear data uncertainty in comparison with other code systems was good.

The work presented is a preliminary verification of the nuclear data sensitivity analysis of responses tallied in fixed-source simulation within OpenMC. Plans to make the method available in the standard release of the OpenMC code are underway. The capability also enables the performance of nuclear data uncertainty analyses in fusion energy systems. The implementation has the advantage of tallying the sensitivity coefficients in one simulation and not having to obtain the adjoint flux in a separate simulation. The nuclear data uncertainties can be propagated in a post-processing step once the cross-section covariances are processed in the energy group of interest. This further eliminates the need to generate random cross sections and run multiple expensive simulations with different sets of perturbed cross sections for uncertainty analysis by TMC.

Future work will include application of the GPT capability to perform sensitivity and uncertainty analyses of nuclear data for the system parameters of fusion

TABLE 4

Uncertainty of the ^{6}Li Tritium Production Rate in the Stacks due to the ^{9}Be Cross Sections

Stack	Reaction	OpenMC (%)	ASUSD (%)	SANDY (%)
1	Total	2.70	2.70	2.78
	Elastic	2.56	2.40	
	$(n, 2n)$	0.85	1.20	1.42
3	Total	2.62	2.60	
	Elastic	2.46	2.40	
	$(n, 2n)$	0.88	1.10	
5	Total	2.30	2.30	
	Elastic	2.12	2.00	
	$(n, 2n)$	0.90	1.10	
7	Total	1.97	2.00	2.03
	Elastic	1.74	1.70	
	$(n, 2n)$	0.91	1.00	1.16

reactor designs, such as the SDR, nuclear heating, energy deposition, and radiation damage.

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Author Contributions

CRediT: **Bamidele Ebiwonjumi:** Conceptualization, Formal analysis, Investigation, Methodology, Validation, Writing – original draft; **Benoit Forget:** Project administration, Supervision, Writing – review & editing; **Ethan Peterson:** Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing.

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References

1. X. Peng et al., "Development of Continuous-Energy Sensitivity Analysis Capability in OpenMC," *Ann. Nucl. Energy*, **110**, 362 (2017); https://doi.org/10.1016/j.anucene.2017.06.061.
2. X. Peng et al., "Calculation of Adjoint-Weighted Reactor Kinetics Parameters in OpenMC," *Ann. Nucl. Energy*, **128**, 231 (2019); https://doi.org/10.1016/j.anucene.2019.01.007.
3. B. C. Kiedrowski and F. B. Brown, "Adjoint-Based k-Eigenvalue Sensitivity Coefficients to Nuclear Data Using Continuous-Energy Monte Carlo," *Nucl. Sci. Eng.*, **174**, *3*, 227 (2013); https://doi.org/10.13182/NSE12-46.
4. C. M. Perfetti, B. T. Rearden, and W. R. Martin, "SCALE Continuous-Energy Eigenvalue Sensitivity Coefficient Calculations," *Nucl. Sci. Eng.*, **182**, *3*, 332 (2016); https://doi.org/10.13182/NSE15-12.
5. M. Aufiero et al., "A Collision History–Based Approach to Sensitivity/Perturbation Calculations in the Continuous Energy Monte Carlo Code SERPENT," *Ann. Nucl. Energy*, **85**, 245 (2015); https://doi.org/10.1016/j.anucene.2015.05.008.
6. C. M. Perfetti and B. T. Rearden, "Development of a Generalized Perturbation Theory Method for Sensitivity Analysis Using Continuous-Energy Monte Carlo Methods," *Nucl. Sci. Eng.*, **182**, *3*, 354 (2016); https://doi.org/10.13182/NSE15-13.
7. J. R. Lamproe et al., "Preliminary Verification of the MCNP Perturbation and Fixed-Source Tally Sensitivity Tools," *Ann. Nucl. Energy*, **194**, 110040 (2023); https://doi.org/10.1016/j.anucene.2023.110040.
8. P. Griveaux et al., "Computation of Sensitivity Coefficients in Fixed Source Simulations with SERPENT2," *Fusion Eng. Des.*, **200**, 114191 (2024); https://doi.org/10.1016/j.fusengdes.2024.114191.
9. C. A. Weaver, C. M. Perfetti, and M. E. Rising, "Fixed Source Sensitivity Calculations for Inertial Confinement Fusion Applications," *Nucl. Sci. Eng.*, **199**, *sup1*, S797 (2025); https://doi.org/10.1080/00295639.2024.2380607.
10. C. Perfetti et al., "Sensitivity Analysis in Coupled Monte Carlo Radiation Transport Simulations," *Nucl. Sci. Eng.*, **198**, *2*, 300 (2024); https://doi.org/10.1080/00295639.2023.2184192.
11. M. Olguin et al., "GEAR-MC and Differential-Operator Methods Applied to Electron-Photon Transport in the Integrated TIGER Series," *Nucl. Sci. Eng.*, **199**, *sup1*, S818 (2025); https://doi.org/10.1080/00295639.2025.2462893.
12. A. A. A. Alhajri, "A Monte Carlo Framework for Nuclear Data Uncertainty Propagation via the Windowed Multipole Formalism," PhD Thesis, Department of Nuclear Science and Engineering, Massachusetts Institute of Technology (2020).
13. B. Sorbom et al., "ARC: A Compact, High-Field, Fusion Nuclear Science Facility and Demonstration Power Plant with Demountable Magnets," *Fusion Eng. Des.*, **100**, 378 (2015); https://doi.org/10.1016/j.fusengdes.2015.07.008.
14. P. Batistoni et al., "Neutronics Experiment on a Helium Cooled Pebble Bed (HCPB) Breeder Blanket Mock-Up," *Fusion Eng. Des.*, **82**, *15–24*, 2095 (2007); https://doi.org/10.1016/j.fusengdes.2007.04.009.
15. C. S. Dunn et al., "Neutronics Analysis of ClLiF: An Alternative Molten-Salt Tritium Breeder," *Fusion Sci. Technol.*, 1 (2025); https://doi.org/10.1080/15361055.2025.2504844.
16. B. Kos, A. Čufar, and I. A. Kodeli, "ASUSD Nuclear Data Sensitivity and Uncertainty Program Package: Validation on Fusion and Fission Benchmark Experiments," *Nucl. Eng. Technol.*, **53**, *7*, 2151 (2021); https://doi.org/10.1016/j.net.2021.01.034.

17. B. C. Kiedrowski and A. Sood, "Prototype Fixed-Source Sensitivity Capability in MCNP6 Applied to Subcritical Thor Core Measurements," *Trans. Am. Nucl. Soc.*, **109**, 853 (2013).

18. M. Williams, "The Relations Between Various Contributon Variables Used in Spatial Channel Theory," *Nucl. Sci. Eng.*, **63**, *2*, 220 (1977); https://doi.org/10.13182/NSE77-A27033.

19. B. Ebiwonjumi, "OpenMC: Sensitivity Analysis Branch," github (2025); https://github.com/Ebiwonjumi/openmc/tree/Sensitivity-Analysis (accessed July 28, 2025).

20. I. A. Kodeli and E. Sartori, "SINBAD—Radiation Shielding Benchmark Experiments," *Ann. Nucl. Energy*, **159**, 108254 (2021); https://doi.org/10.1016/j.anucene.2021.108254.

21. A. Trkov et al., "IRDFF-II: A New Neutron Metrology Library," *Nucl. Data Sheets*, **163**, 1 (2020); https://doi.org/10.1016/j.nds.2019.12.001.

22. T. D. Bohm and B. A. Lindley, "Initial Neutronics Investigation of a Chlorine Salt–Based Breeder Blanket," *Fusion Sci. Technol.*, **79**, *8*, 995 (2023); https://doi.org/10.1080/15361055.2022.2136923.

23. T. Taylor et al., "Sensitivity to Chlorine Nuclear Data in Molten Chloride Fast Reactors," presented at the *41st Annual Conf. of the Canadian Nuclear Society and the 46th Annual CNS/CNA Student Conf.* Virtual Mtg. 2022, Canadian Nuclear Society.

24. K. Hanselman et al., "Improved Modeling of Neutron-Induced Reactions on Chlorine Isotopes Aided Through New (n, p) and (n, α) Measurements at LANSCE," *Phys. Rev. C.*, **110**, *2*, 024609 (2024); https://doi.org/10.1103/PhysRevC.110.024609.

25. E. Nunnenmann, U. Fischer, and R. Stieglitz, "Sensitivity and Uncertainty Analysis for the Tritium Breeding Ratio of a DEMO Fusion Reactor with a Helium Cooled Pebble Bed Blanket," *EPJ Web Conf.*, **146**, 09025 (2017).

26. R. MacFarlane et al., "The NJOY Nuclear Data Processing System, Version 2016," Los Alamos National Laboratory (2017).

27. L. Fiorito et al., "Nuclear Data Uncertainty Propagation to Integral Responses Using SANDY," *Ann. Nucl. Energy*, **101**, 359 (2017); https://doi.org/10.1016/j.anucene.2016.11.026.

28. D. Leichtle et al., "Sensitivity and Uncertainty Analyses of the Tritium Production in the HCPB Breeder Blanket Mock-Up Experiment," *Fusion Eng. Des.*, **82**, *15–24*, 2406 (2007); https://doi.org/10.1016/j.fusengdes.2007.08.007.

29. R. Perel, J. Wagschal, and Y. Yeivin, "Monte Carlo Calculation of Point-Detector Sensitivities to Material Parameters," *Nucl. Sci. Eng.*, **124**, *1*, 197 (1996); https://doi.org/10.13182/NSE96-A24235.

30. I. Kodeli, "Multidimensional Deterministic Nuclear Data Sensitivity and Uncertainty Code System: Method and Application," *Nucl. Sci. Eng.*, **138**, *1*, 45 (2001); https://doi.org/10.13182/NSE00-43.

31. T. M. Evans et al., "Denovo: A New Three-Dimensional Parallel Discrete Ordinates Code in SCALE," *Nucl. Technol.*, **171**, *2*, 171 (2010); https://doi.org/10.13182/NT171-171.

32. B. Kos, Baker Hughes, ASUSD Results - Nuclear Engineering Technology Paper, Personal Communication (Aug. 28, 2025).

33. A. Alhajri et al., "The Windowed Multipole Formalism and Applications to Uncertainty Quantification," *Ann. Nucl. Energy*, **156**, 108168 (2021); https://doi.org/10.1016/j.anucene.2021.108168.